LEARNMATE – AI STUDYBUDDY

4. Project Planning Phase

Organizing the project into setup, backend, database, AI, execution and testing activities.

Date	21-09-26
Team ID	SWTID-2026-5714
Project Name	LearnMate - AI StudyBuddy
Maximum marks	4 Marks



AI-powered learning assistance and smart study support platform

Planned Work Sequence

Key Points

- Create the AI StudyBuddy project folder and initialize the Node.js project.
- Install Express, Mongoose, bcryptjs, jsonwebtoken, cors, dotenv, multer and @google/genai.
- Configure environment variables for PORT, MongoDB, JWT and Gemini API access.
- Create the modular backend structure with routes, controllers, middleware, models and utilities.



Reference visual from the supplied LearnMate project document.

- Configure MongoDB and Mongoose schemas.
- Integrate Gemini AI services and file-upload middleware.
- Execute the backend locally and test REST APIs using Postman or Thunder Client.

Planned Modules

Key Points

- Authentication and authorization.
- Study material management.
- AI summary, flashcard, quiz and study-plan generation.
- Database management.
- Administrative APIs and monitoring.

Execution Readiness

Key Points

- MongoDB should be running and required environment variables must be configured before starting the backend.
- The development server is started using npm start and APIs are verified through testing tools.